

Lista 9- Cálculo II Jeann Victor Batista

① $z = x^3 + y^3 + xy, x = \sin t, y = e^t$

$$z = \sin^3 t + e^{3t} + \sin t \cdot e^t = (\sin t)^3 + e^{3t} + \sin t \cdot e^t$$

$$\frac{dz}{dt} = 3 \cdot \sin^2 t \cdot \cos t + e^{3t} \cdot 3 + \sin t \cdot e^t + \cos t \cdot e^t$$

b) $z = \cos(x + y), x = 5t^3, y = 11t$

$$z = \cos(5t^3 + 11t)$$

$$\frac{dz}{dt} = -\sin(5t^3 + 11t) \cdot (15t^2 + 11)$$

c) $w = x \cdot e^{y/z}, x = t^2, y = 1-t, z = \frac{1}{1+t} + 2t$

$$w = t^2 \cdot e^{(1-t)/(1+t)} + t^2 \cdot \left(\frac{-3}{(1+t)^2} \cdot e^{(1-t)/(1+t)} \right)$$

$$\left(e^{(1-t)/(1+t)} \right)' = e^{(1-t)/(1+t)} \cdot \left(\frac{(1+t) \cdot (-1) - (1-t) \cdot 1}{(1+t)^2} \right) = \frac{1-t-1+t}{(1+t)^2} = \frac{-2}{(1+t)^2} \cdot e^{(1-t)/(1+t)}$$

d) $w = \ln \sqrt{x^2 + y^2 + z^2}, x = \sin t, y = \cos t, z = \tan t$

$$w = \ln \sqrt{\sin^2 t + \cos^2 t + \tan^2 t}$$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

$$\frac{\partial w}{\partial x} = \frac{1}{2} \cdot \frac{2x}{x^2 + y^2 + z^2} = \frac{x}{x^2 + y^2 + z^2} \quad \frac{dx}{dt} = \cos t$$

$$\frac{\partial w}{\partial y} = \frac{1}{2} \cdot \frac{2y}{x^2 + y^2 + z^2} = \frac{y}{x^2 + y^2 + z^2} \quad \frac{dy}{dt} = -\sin t$$

$$\frac{\partial w}{\partial z} = \frac{1}{2} \cdot \frac{2z}{x^2 + y^2 + z^2} = \frac{z}{x^2 + y^2 + z^2} \quad \frac{dz}{dt} = \sec^2 t$$

$$\frac{dw}{dt} = \left(\frac{x}{x^2 + y^2 + z^2} \right) \cdot \cos t - \left(\frac{y}{x^2 + y^2 + z^2} \right) \cdot \sin t + \left(\frac{z}{x^2 + y^2 + z^2} \right) \cdot \sec^2 t$$

$$\frac{dw}{dt} = \frac{\sin t \cdot \cos t}{\sec^2 t} - \frac{\cos t \cdot \sin t}{\sec^2 t} + \frac{\tan t \cdot \sec^2 t}{\sec^2 t} = \tan t$$



2

$$a) \frac{\partial Z}{\partial S} = \frac{\partial Z}{\partial X} \cdot \frac{\partial X}{\partial S} + \frac{\partial Z}{\partial Y} \cdot \frac{\partial Y}{\partial S} \quad Z = X^2 Y^3 \quad X = S \cdot \text{cost} \quad Y = S \cdot \text{rent}$$

$$\frac{\partial Z}{\partial X} = 2XY^3 \quad \frac{\partial X}{\partial S} = \text{cost} \quad \frac{\partial Z}{\partial Y} = X^2 \cdot 3Y^2 \quad \frac{\partial Y}{\partial S} = \text{rent}$$

$$\frac{\partial Z}{\partial S} = 2XY^3 \cdot \text{cost} + 3X^2Y^2 \cdot \text{rent}$$

$$\frac{\partial Z}{\partial S} = 2(S \cdot \text{cost}) (S \cdot \text{rent})^3 \cdot \text{cost} + 3(S \cdot \text{cost})^2 (S \cdot \text{rent})^2 \cdot \text{rent}$$

$$\frac{\partial Z}{\partial S} = 2S \cdot \text{cost} \cdot S^3 \cdot \text{rent}^3 \cdot \text{cost} + 3S^2 \cdot \text{cost}^2 \cdot S^2 \cdot \text{rent}^2 \cdot \text{rent}$$

$$\frac{\partial Z}{\partial S} = 2S^4 \cdot \text{cost}^2 \cdot \text{rent}^3 + 3S^4 \cdot \text{cost}^2 \cdot \text{rent}^3$$

$$\frac{\partial Z}{\partial S} = 2S^4 \cdot \text{cost}^2 \cdot \text{rent}^3 + 3S^4 \cdot \text{cost}^2 \cdot \text{rent}^3$$

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$$\frac{\partial Z}{\partial S} = 2S^4 \cdot \text{cost}^2 \cdot \text{rent}^3 + 3S^4 \cdot \text{cost}^2 \cdot \text{rent}^3$$

b)

$$z = \sin \theta \cdot \cos \phi, \quad \theta = st, \quad \phi = s^2 t$$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial \theta} \cdot \frac{\partial \theta}{\partial s} + \frac{\partial z}{\partial \phi} \cdot \frac{\partial \phi}{\partial s}$$

$$\frac{\partial z}{\partial s} = \cos \theta \cdot \cos \phi \cdot \frac{\partial \theta}{\partial s} + t^2 \frac{\partial z}{\partial \theta} - \sin \theta \cdot \sin \phi \cdot \frac{\partial \phi}{\partial s} = \cos \theta \cdot \cos \phi \cdot t^2 - \sin \theta \cdot \sin \phi \cdot 2st$$

$$\frac{\partial z}{\partial s} = \cos(st) \cdot \cos(s^2 t) \cdot t^2 - \sin(st) \cdot \sin(s^2 t) \cdot 2st$$

$$\frac{\partial z}{\partial s} = \cos(st^3) \cdot \cos(s^2 t) \cdot t^2 - \sin(st^3) \cdot \sin(s^2 t) \cdot 2st$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial \theta} \cdot \frac{\partial \theta}{\partial t} + \frac{\partial z}{\partial \phi} \cdot \frac{\partial \phi}{\partial t}$$

$$\frac{\partial z}{\partial t} = \cos \theta \cdot \cos \phi \cdot \frac{\partial \theta}{\partial t} - s \cdot \sin \theta \cdot \sin \phi \cdot \frac{\partial \phi}{\partial t} = \cos(st) \cdot \cos(s^2 t) \cdot s - \sin(st) \cdot \sin(s^2 t) \cdot 2st$$

$$\frac{\partial z}{\partial t} = \cos(st) \cdot \cos(s^2 t) \cdot s - \sin(st) \cdot \sin(s^2 t) \cdot 2st$$

$$\frac{\partial z}{\partial t} = \cos(st^3) \cdot \cos(s^2 t) \cdot s - \sin(st^3) \cdot \sin(s^2 t) \cdot 2st$$

$$c) \quad z = e^r \cdot \cos \theta, \quad r = st, \quad \theta = -\sqrt{s^2 + t^2}$$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial r} \cdot \frac{\partial r}{\partial s} + \frac{\partial z}{\partial \theta} \cdot \frac{\partial \theta}{\partial s}$$

$$\frac{\partial z}{\partial r} = e^r \cdot \cos \theta \cdot \frac{\partial r}{\partial s} = \frac{\partial z}{\partial \theta} = -e^r \cdot \sin \theta \cdot \frac{\partial \theta}{\partial s} = 1 \cdot (s^2 + t^2)^{-1/2} \cdot s = \frac{s}{\sqrt{s^2 + t^2}}$$

$$\frac{\partial z}{\partial r} = e^r \cdot \cos \theta \cdot t - e^r \cdot \sin \theta \cdot \frac{s}{\sqrt{s^2 + t^2}} = e^{st} \cdot \cos(-\sqrt{s^2 + t^2}) \cdot t - e^{st} \cdot \sin(-\sqrt{s^2 + t^2}) \cdot \frac{s}{\sqrt{s^2 + t^2}}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial r} \cdot \frac{\partial r}{\partial t} + \frac{\partial z}{\partial \theta} \cdot \frac{\partial \theta}{\partial t}$$

$$\frac{\partial z}{\partial r} = e^r \cdot \cos \theta \cdot \frac{\partial r}{\partial t} = s \cdot \frac{\partial z}{\partial \theta} = -e^r \cdot \sin \theta \cdot \frac{\partial \theta}{\partial t} = -e^{st} \cdot \sin(-\sqrt{s^2 + t^2}) \cdot \frac{t}{\sqrt{s^2 + t^2}}$$

$$\frac{\partial z}{\partial t} = e^r \cdot \cos \theta \cdot s - e^r \cdot \sin \theta \cdot \frac{t}{\sqrt{s^2 + t^2}} = e^{st} \cdot \cos(\sqrt{s^2 + t^2}) \cdot s - e^{st} \cdot \sin(\sqrt{s^2 + t^2}) \cdot \frac{t}{\sqrt{s^2 + t^2}}$$

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$$d) z = \tan\left(\frac{u}{v}\right), u = 2s + 3t, v = 3s - 2t$$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial s} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial s}$$

$$\frac{\partial z}{\partial u} = \sec^2\left(\frac{u}{v}\right) \cdot \frac{1}{v} \quad \frac{\partial u}{\partial s} = 2 \quad \frac{\partial z}{\partial v} = \sec^2\left(\frac{u}{v}\right) \cdot \left(-\frac{u}{v^2}\right) \quad \frac{\partial v}{\partial s} = 3$$

$$\frac{\partial z}{\partial s} = \sec^2\left(\frac{u}{v}\right) \cdot \frac{1}{v} = 2 + \sec^2\left(\frac{u}{v}\right) \cdot \left(-\frac{u}{v^2}\right) \cdot 3$$

$$\frac{\partial z}{\partial s} = \sec^2\left(\frac{2s+3t}{3s-2t}\right) \cdot \frac{1}{3s-2t} - 2 + \sec^2\left(\frac{2s+3t}{3s-2t}\right) \cdot \left(\frac{-2s-3t}{(3s-2t)^2}\right) \cdot 3$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial t} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial t}$$

$$\frac{\partial z}{\partial v} = \sec^2\left(\frac{u}{v}\right) \cdot \frac{1}{v} \quad \frac{\partial u}{\partial t} = 3 \quad \frac{\partial z}{\partial v} = \sec^2\left(\frac{u}{v}\right) \cdot \left(-\frac{u}{v^2}\right) \quad \frac{\partial v}{\partial t} = -2$$

$$\frac{\partial z}{\partial t} = \sec^2\left(\frac{u}{v}\right) \cdot \frac{1}{v} \cdot 3 + \sec^2\left(\frac{u}{v}\right) \cdot \left(-\frac{u}{v^2}\right) \cdot (-2)$$

$$\frac{\partial z}{\partial t} = \sec^2\left(\frac{2s+3t}{3s-2t}\right) \cdot \frac{1}{3s-2t} + 3 + \sec^2\left(\frac{2s+3t}{3s-2t}\right) \cdot \left(\frac{-2s-3t}{(3s-2t)^2}\right) \cdot (-2)$$

$$③ a) z = x^2 + xy^3, x = uv^2 + w^3, y = u + ve^w \quad u=2, v=1, w=0$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial x} = 2x + y^3 = 2 \cdot (uv^2 + w^3) + (u + ve^w)^3 = 2 \cdot (2 \cdot 1^2 + 0^3) + (2 + 1 \cdot e^0)^3 = 2 \cdot 2 + 27 = 31$$

$$\frac{\partial x}{\partial u} = v^2 = 1^2 = 1 \quad \frac{\partial z}{\partial y} = 3y^2 x = 3 \cdot (u + ve^w)^2 \cdot (uv^2 + w^3) = 3 \cdot (2 + 1 \cdot e^0)^2 \cdot 2 \cdot 1^2 + 0 = 27 \cdot 2 = 54$$

$$\frac{\partial y}{\partial u} = 1$$

$$\frac{\partial z}{\partial u} = 31 \cdot 1 + 54 \cdot 1 = 85$$

$$\frac{\partial z}{\partial u}$$



$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$

$$\frac{\partial z}{\partial x} = 31 \quad \frac{\partial x}{\partial v} = 2 \cdot 2 \cdot 1 = 1 \quad \frac{\partial z}{\partial y} = 54 \quad \frac{\partial y}{\partial v} = e^w - e^0 = 1$$

$$\frac{\partial z}{\partial v} = 31 \cdot 4 + 54 \cdot 1 = 124 + 54 = 178$$

$$\frac{\partial z}{\partial w} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial w} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial w}$$

$$\frac{\partial z}{\partial x} = 31 \quad \frac{\partial x}{\partial w} = 3w^2 = 3 \cdot 0^2 = 0 \quad \frac{\partial z}{\partial y} = 54 \quad \frac{\partial y}{\partial w} = v e^w - 1 \cdot e^0 = 1$$

$$\frac{\partial z}{\partial w} = 31 \cdot 0 + 54 \cdot 1 = 54$$

b) $u = \sqrt{r^2 + s^2}$, $r = y + x \cdot \cos t$, $s = x + y \cdot \sin t$ $x=1, y=2, t=0$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x}$$

$$\frac{\partial u}{\partial r} = \frac{1}{2} \cdot (r^2 + s^2)^{-1/2} \cdot 2r = \frac{r}{\sqrt{r^2 + s^2}} = \frac{y + x \cdot \cos t}{\sqrt{(y + x \cdot \cos t)^2 + (x + y \cdot \sin t)^2}} = \frac{2 + 1 \cdot \cos 0}{\sqrt{(2 + 1 \cdot \cos 0)^2 + (1 + 2 \sin 0)^2}}$$

$$\frac{\partial u}{\partial r} = \frac{2+1}{\sqrt{(2+1)^2 + 1}} = \frac{3}{\sqrt{10}} \quad \frac{\partial r}{\partial x} = \cos t = \cos 0 = 1$$

$$\frac{\partial u}{\partial s} = \frac{s}{\sqrt{r^2 + s^2}} = \frac{x + y \cdot \sin t}{\sqrt{r^2 + s^2}} = \frac{1}{\sqrt{10}} \quad \frac{\partial s}{\partial x} = 1$$

$$\frac{\partial u}{\partial x} = \frac{3}{\sqrt{10}} \cdot 1 + \frac{1}{\sqrt{10}} \cdot 1 = \frac{4}{\sqrt{10}}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y}$$

$$\frac{\partial u}{\partial r} = \frac{3}{\sqrt{10}} \quad \frac{\partial r}{\partial y} = 1 \quad \frac{\partial u}{\partial s} = \frac{1}{\sqrt{10}} \quad \frac{\partial s}{\partial y} = \sin t = \sin 0 = 0$$

$$\frac{\partial u}{\partial y} = \frac{3}{\sqrt{10}} \cdot 1 + \frac{1}{\sqrt{10}} \cdot 0 = \frac{3}{\sqrt{10}}$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial t} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial t}$$



$$\frac{\partial u}{\partial r} = \frac{3}{\sqrt{10}} \quad \frac{\partial r}{\partial t} = -x \cdot \sin \theta = -1 \cdot \sin \theta = 0$$

$$\frac{\partial u}{\partial s} = \frac{1}{\sqrt{10}} \quad \frac{\partial s}{\partial t} = y \cdot \cos \theta = 2 \cdot \cos \theta = 2$$

$$\frac{\partial u}{\partial t} = \frac{3}{\sqrt{10}} \cdot 0 + \frac{1}{\sqrt{10}} \cdot 2 = \frac{2}{\sqrt{10}}$$

$$C) N = (P+Q)/(P+R), p = u+vw, q = v+uw, r = w+uv \quad u=2, v=3, w=4$$

$$\frac{\partial N}{\partial v} = \frac{\partial N}{\partial p} \cdot \frac{\partial p}{\partial v} + \frac{\partial N}{\partial q} \cdot \frac{\partial q}{\partial v} + \frac{\partial N}{\partial r} \cdot \frac{\partial r}{\partial v}$$

$$\frac{\partial N}{\partial p} = \frac{p+r-p-q}{(p+r)^2} = \frac{r-q}{(p+r)^2} = \frac{w+uv-v-uw}{(u+vw+w+uv)^2} = \frac{4+2 \cdot 3-3-2 \cdot 4}{(2+3 \cdot 4+4+2 \cdot 3)^2} = \frac{-1}{576}$$

$$\frac{\partial p}{\partial v} = 1 \quad \frac{\partial N}{\partial q} = \frac{p+r-(p+q) \cdot 0}{(p+r)^2} = \frac{u+vw+w+uv}{576} = \frac{2+3 \cdot 4+4+2 \cdot 3}{576} = \frac{24}{576} = \frac{1}{24}$$

$$\frac{\partial q}{\partial v} = w=4 \quad \frac{\partial N}{\partial r} = \frac{0-(p+q) \cdot 1}{(p+r)^2} = \frac{-p-q}{(p+r)^2} = \frac{-u-vw-v-uw}{576} = \frac{-2-3 \cdot 4-3-2 \cdot 4}{576} = \frac{-25}{576}$$

$$\frac{\partial r}{\partial v} = v=3$$

$$\frac{\partial N}{\partial v} = \frac{-1}{576} \cdot 1 + \frac{1}{24} \cdot 4 - \frac{25}{576} \cdot 3 = \frac{-1}{576} + \frac{4}{6} - \frac{75}{576} = \frac{-76}{576} + \frac{4}{6} = \frac{20}{576} = \frac{5}{144}$$

$$\frac{\partial N}{\partial v} = \frac{\partial N}{\partial p} \cdot \frac{\partial p}{\partial v} + \frac{\partial N}{\partial q} \cdot \frac{\partial q}{\partial v} + \frac{\partial N}{\partial r} \cdot \frac{\partial r}{\partial v}$$

$$\frac{\partial N}{\partial p} = \frac{-1}{576} \quad \frac{\partial p}{\partial v} = w=4 \quad \frac{\partial N}{\partial q} = \frac{1}{576} \quad \frac{\partial q}{\partial v} = 4 \quad \frac{\partial N}{\partial r} = \frac{-25}{576} \quad \frac{\partial r}{\partial v} = u=2$$

$$\frac{\partial N}{\partial v} = \frac{-1}{576} \cdot 4 + \frac{1}{576} \cdot 16 - \frac{25}{576} \cdot 2 = \frac{-4}{576} + \frac{16}{576} - \frac{50}{576} = \frac{-8}{576} = \frac{-1}{72}$$

$$\frac{\partial N}{\partial w} = \frac{\partial N}{\partial p} \cdot \frac{\partial p}{\partial w} + \frac{\partial N}{\partial q} \cdot \frac{\partial q}{\partial w} + \frac{\partial N}{\partial r} \cdot \frac{\partial r}{\partial w} = \frac{-1}{576} \cdot 3 + \frac{1}{576} \cdot 2 - \frac{25}{576} \cdot 1 = \frac{-3}{576} + \frac{2}{576} - \frac{25}{576} = \frac{-26}{576} = \frac{-13}{288}$$

$$\frac{\partial N}{\partial w} = \frac{-1}{576} \cdot 3 + \frac{1}{576} \cdot 2 - \frac{25}{576} \cdot 1 = \frac{-3}{576} + \frac{2}{576} - \frac{25}{576} = \frac{-26}{576} = \frac{-13}{288}$$

tilibra

2) a) $z^3 - xy + yz + y^3 - 2 = 0 \rightarrow F, P = (1, 1, 1)$

$$\frac{\partial z}{\partial x} = \frac{-\partial F / \partial x}{\partial F / \partial z} \quad \frac{\partial F}{\partial x} = -y \quad \frac{\partial F}{\partial z} = 3z^2 + y \quad \frac{\partial z}{\partial x} = \frac{-y}{3z^2 + y}$$

$$\frac{\partial z}{\partial y} = \frac{-\partial F / \partial y}{\partial F / \partial z} \quad \frac{\partial F}{\partial y} = -x + z + 3y^2 \quad \frac{\partial F}{\partial z} = 3z^2 + y \quad \frac{\partial z}{\partial y} = \frac{-x - z - 3y^2}{3z^2 + y}$$

$$\frac{\partial z}{\partial x} = \frac{1}{3 \cdot 1^2 + 1} = \frac{1}{4}$$

$$\frac{\partial z}{\partial y} = \frac{1 - 1 - 3 \cdot 1^2}{3 \cdot 1^2 + 1} = \frac{-3}{4}$$

b) $1/x + 1/y + 1/z - 1 = 0 \rightarrow F, P = (2, 3, 6) \quad F = x^{-1} + y^{-1} + z^{-1} - 1 = 0$

$$\frac{\partial z}{\partial x} = \frac{-\partial F / \partial x}{\partial F / \partial z} \quad \frac{\partial F}{\partial x} = -\frac{1}{x^2} = \frac{-1}{2^2} \quad \frac{\partial F}{\partial z} = \frac{-1}{z^2}$$

$$\frac{\partial z}{\partial x} = \frac{1}{2^2} \bigg/ \frac{-1}{6^2} = \frac{1}{4} \cdot \frac{36}{-1} = -9$$

$$\frac{\partial z}{\partial y} = \frac{-\partial F / \partial y}{\partial F / \partial z} \quad \frac{\partial F}{\partial y} = \frac{-1}{y^2} \quad \frac{\partial F}{\partial z} = \frac{-1}{z^2}$$

$$\frac{\partial z}{\partial y} = \frac{1}{3^2} \bigg/ \frac{-1}{6^2} = \frac{1}{9} \cdot \frac{36}{-1} = -4$$

c) $\sin(x+y) + \sin(y+z) + \sin(x+z) = 0 \rightarrow F, P = (\pi, \pi, \pi)$

$$\frac{\partial z}{\partial x} = \frac{-\partial F / \partial x}{\partial F / \partial z} \quad \frac{\partial F}{\partial x} = \cos(x+y) + \cos(x+z) \quad \frac{\partial F}{\partial z} = \cos(y+z) + \cos(x+z)$$

$$\frac{\partial z}{\partial x} = \frac{-\cos(x+y) - \cos(x+z)}{\cos(y+z) + \cos(x+z)} = \frac{-\cos(2\pi) - \cos(2\pi)}{\cos(2\pi) + \cos(2\pi)} = \frac{-1 - 1}{1 + 1} = -1$$

$$\frac{\partial z}{\partial y} = \frac{-\partial F / \partial y}{\partial F / \partial z} \quad \frac{\partial F}{\partial y} = \cos(x+y) + \cos(y+z) \quad \frac{\partial F}{\partial z} = \cos(y+z) + \cos(x+z)$$

$$\frac{\partial z}{\partial y} = \frac{-\cos(x+y) - \cos(y+z)}{\cos(y+z) + \cos(x+z)} = \frac{-\cos(2\pi) - \cos(2\pi)}{\cos(2\pi) + \cos(2\pi)} = \frac{-1 - 1}{1 + 1} = -1$$

$$(5) \frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = 5x(g(t), h(t)) \cdot g'(t) + f_y(g(t), h(t)) \cdot h'(t)$$

$$\frac{dx}{dt} = g'(3) = 5, \quad \frac{dy}{dt} = h'(3) = -4, \quad f_x(2,7) = 6, \quad f_y(2,7) = -8$$

$$\frac{dz}{dt} = f_x(2,7) \cdot g'(3) + f_y(2,7) \cdot h'(3) = 6 \cdot 5 - 8 \cdot (-4) = 62$$

$$t=3$$

$$(6) a) \frac{\partial z}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r} \quad x = r \cdot \cos \theta \quad y = r \cdot \sin \theta$$

$$\frac{\partial x}{\partial r} = \cos \theta \quad \frac{\partial y}{\partial r} = \sin \theta$$

$$\frac{\partial z}{\partial r} = \frac{\partial f}{\partial x} \cos \theta + \frac{\partial f}{\partial y} \sin \theta$$

$$\frac{\partial z}{\partial \theta} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta} \quad \left\| \frac{\partial x}{\partial \theta} = -r \cdot \sin \theta \quad \frac{\partial y}{\partial \theta} = r \cdot \cos \theta \right.$$

$$\frac{\partial z}{\partial \theta} = \frac{\partial f}{\partial x} (-r \cdot \sin \theta) + \frac{\partial f}{\partial y} (r \cdot \cos \theta)$$

$$b) \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 = \left(\frac{\partial z}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta} \right)^2$$

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial r} = \frac{\partial z}{\partial x} \cos \theta + \frac{\partial z}{\partial y} \sin \theta$$

$$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial \theta} = \frac{\partial z}{\partial x} (-r \cdot \sin \theta) + \frac{\partial z}{\partial y} (r \cdot \cos \theta)$$

$$\left(\frac{\partial z}{\partial r} \right)^2 = \left(\frac{\partial z}{\partial x} \cos \theta + \frac{\partial z}{\partial y} \sin \theta \right)^2$$

$$\left(\frac{\partial z}{\partial x} \right)^2 \cos^2 \theta + 2 \cdot \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} \cos \theta \sin \theta + \left(\frac{\partial z}{\partial y} \right)^2 \sin^2 \theta$$

$$\boxed{\text{tilibra}} \quad \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta} \right)^2 = \frac{1}{r^2} \left(-r \cdot \frac{\partial z}{\partial x} \sin \theta + r \cdot \frac{\partial z}{\partial y} \cos \theta \right)^2$$

$$\left(\frac{-az}{ax} \sin \theta + \frac{az}{ay} \cos \theta \right)^2 \rightarrow \left(\frac{az}{ax} \right)^2 \sin^2 \theta - 2 \cdot \frac{az}{ax} \cdot \frac{az}{ay} \sin \theta \cos \theta + \left(\frac{az}{ay} \right)^2 \cos^2 \theta$$

$$\left(\frac{az}{ax} \right)^2 + \frac{1}{r^2} \left(\frac{az}{ay} \right)^2$$

$$\left[\left(\frac{az}{ax} \right)^2 \cos^2 \theta + \left(\frac{az}{ay} \right)^2 \sin^2 \theta \right] + \left[\left(\frac{az}{ax} \right)^2 \sin^2 \theta + \left(\frac{az}{ay} \right)^2 \cos^2 \theta \right] + 2 \cdot \frac{az}{ax} \cdot \frac{az}{ay} \sin \theta \cos \theta$$

$$\sin \theta - 2 \cdot \frac{az}{ax} \cdot \frac{az}{ay} \sin \theta \cos \theta$$

$$\left(\frac{az}{ax} \right)^2 (\cos^2 \theta + \sin^2 \theta) + \left(\frac{az}{ay} \right)^2 (\sin^2 \theta + \cos^2 \theta)$$

$$\left(\frac{az}{ax} \right)^2 + \left(\frac{az}{ay} \right)^2$$

7) a) $f(x,y) = \ln(x^2 + y^2)$, $P(1,1)$

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

$$\frac{\partial f}{\partial x} = \frac{1}{x^2 + y^2} \cdot 2x = \frac{2x}{x^2 + y^2} \quad \frac{\partial f}{\partial y} = \frac{1}{x^2 + y^2} \cdot 2y = \frac{2y}{x^2 + y^2}$$

$$\frac{\partial f}{\partial x} \rightarrow \frac{2 \cdot 1}{1^2 + 1^2} = 1 \quad \frac{\partial f}{\partial y} = \frac{2 \cdot 1}{1^2 + 1^2} = 1 \quad \nabla f = (1,1)$$

b) $f(x,y) = xy^2$, $P(2,-2)$

$$\frac{\partial f}{\partial x} = y^2 \quad \frac{\partial f}{\partial y} = 2xy \quad \left(\frac{\partial f}{\partial x} \rightarrow (-1)^2 = 1 \quad \frac{\partial f}{\partial y} = 2 \cdot 2 \cdot (-1) = -4 \right) \quad \nabla f = (1, -4)$$

c) $f(x,y) = \sqrt{2x+3y}$, $P(-1,2)$

$$\frac{\partial f}{\partial x} = \frac{1}{2} \cdot (2x+3y)^{-1/2} \cdot 2 = \frac{1}{\sqrt{2x+3y}} \rightarrow \frac{3}{2 \cdot \sqrt{2}} = \frac{3}{2 \cdot 2} = \frac{3}{4} \quad \nabla f = \left(\frac{1}{2}, \frac{3}{4} \right)$$

d) $f(x,y,z) = x^3 + y^3 - 2z^2 + 2 \ln x, P = (1,1,1)$

$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right), \frac{\partial f}{\partial x} = 2x + 2 \cdot \frac{1}{x} \rightarrow 2 \cdot 1 + 2 \cdot \frac{1}{1} = 2 + 2 = 4$

$\frac{\partial f}{\partial y} = 2y \rightarrow 2 \cdot 1 = 2$

$\frac{\partial f}{\partial z}$

$\frac{\partial f}{\partial z} = -4z + \ln x \rightarrow -4 \cdot 1 + \ln 1 = -4 + 0 = -4$

$\frac{\partial f}{\partial z}$

$f = (3, 2, -4)$

e) $f(x,y,z) = (x^2 + y^2 + z^2)^{-1/2} + \ln(xyz), P = (-1, 2, -2)$

$\frac{\partial f}{\partial x} = -\frac{1}{2} \cdot (x^2 + y^2 + z^2)^{-3/2} \cdot 2x + \frac{1}{xyz} \cdot yz = \frac{-x}{\sqrt{(x^2 + y^2 + z^2)^3}} + \frac{1}{x}$

$\frac{\partial f}{\partial y} = -\frac{1}{2} \cdot (x^2 + y^2 + z^2)^{-3/2} \cdot 2y + \frac{1}{xyz} \cdot xz = \frac{-y}{\sqrt{(x^2 + y^2 + z^2)^3}} + \frac{1}{y}$

$\frac{\partial f}{\partial z} = -\frac{1}{2} \cdot (x^2 + y^2 + z^2)^{-3/2} \cdot 2z + \frac{1}{xyz} \cdot xy = \frac{-z}{\sqrt{(x^2 + y^2 + z^2)^3}} + \frac{1}{z}$

$\frac{\partial f}{\partial x} \rightarrow \frac{1}{\sqrt{((-1)^2 + 2^2 + (-2)^2)^3}} - \frac{1}{1} = \frac{1}{\sqrt{9^3}} - \frac{1}{1} = \frac{1}{27} - 1 = \frac{1}{27} - \frac{27}{27} = \frac{-26}{27}$

$\frac{\partial f}{\partial y} \rightarrow \frac{-2}{27} + \frac{1}{2} = \frac{-4 + 27}{54} = \frac{23}{54}$

$\frac{\partial f}{\partial z} \rightarrow \frac{2}{27} - \frac{1}{2} = \frac{4 - 27}{54} = \frac{-23}{54}$

$\nabla f = \left(\frac{-26}{27}, \frac{23}{54}, \frac{-23}{54} \right)$

8) a) $f(x,y) = 2xy - 3y^2, P = (5,5), u = 4i + 3j$

$\frac{\partial f}{\partial x} = 2y \rightarrow 2 \cdot 5 = 10$
 $\frac{\partial f}{\partial y} = 2x - 6y \rightarrow 2 \cdot 5 - 6 \cdot 5 = -20$

$\nabla f = (10, -20)$

$\|u\| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5 \rightarrow u = \left(\frac{4}{5}, \frac{3}{5} \right)$

$D_u f(5,5) = 10 \cdot \frac{4}{5} - 20 \cdot \frac{3}{5} = 8 - 12 = -4$

$$b) f(x,y) = 2x^3 + y^2, P = (-1,1), \mu = 3i - 4j$$

$$\frac{\partial f}{\partial x} = 6x^2 \rightarrow 6 \cdot (-1) = -6 \quad \frac{\partial f}{\partial y} = 2y \rightarrow 2 \cdot 1 = 2$$

$$\|u\| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5 \rightarrow \mu = \left(\frac{3}{5}, -\frac{4}{5}\right)$$

$$\nabla f = (-6, 2)$$

$$D_u f(3, -4) = -6 \cdot \frac{3}{5} + 2 \cdot \left(-\frac{4}{5}\right) = \frac{-18}{5} - \frac{8}{5} = \frac{-26}{5} = -5.2$$

$$c) f(x,y) = x/x^2 + y^2, P = (1,2), \mu = (3,5)$$

$$\frac{\partial f}{\partial x} = \frac{(x^2 + y^2) \cdot 1 - x \cdot 2x}{(x^2 + y^2)^2} = \frac{x^2 + y^2 - 2x^2}{(x^2 + y^2)^2} = \frac{-x^2 + y^2}{(x^2 + y^2)^2} \rightarrow \frac{-1^2 + 2^2}{(1^2 + 2^2)^2} = \frac{-1 + 4}{25} = \frac{3}{25}$$

$$\frac{\partial f}{\partial y} = \frac{(x^2 + y^2) \cdot 0 - x \cdot 2y}{(x^2 + y^2)^2} = \frac{-2xy}{(x^2 + y^2)^2} \rightarrow \frac{-2 \cdot 1 \cdot 2}{(1^2 + 2^2)^2} = \frac{-4}{25}$$

$$\nabla f = \left(\frac{3}{25}, -\frac{4}{25}\right)$$

$$\|u\| = \sqrt{3^2 + 5^2} = \sqrt{34} \rightarrow \mu = \left(\frac{3}{\sqrt{34}}, \frac{5}{\sqrt{34}}\right)$$

$$D_u f(3,5) = \frac{3}{25} \cdot \frac{3}{\sqrt{34}} - \frac{4}{25} \cdot \frac{5}{\sqrt{34}} = \frac{9}{25\sqrt{34}} - \frac{20}{25\sqrt{34}} = \frac{-11}{25\sqrt{34}}$$

$$d) f(x,y,z) = xy + yz + zx, P = (1,1,2), \mu = 3i + 6j + 2k$$

$$\frac{\partial f}{\partial x} = y + z \rightarrow -1 + 2 = 1 \quad \nabla f = (1, 3, 0)$$

$$\frac{\partial f}{\partial y} = x + z$$

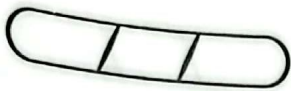
$$\frac{\partial f}{\partial z} = x + y \rightarrow 1 + 1 = 2 \quad \|u\| = \sqrt{3^2 + 6^2 + 2^2} = \sqrt{49} = 7 \rightarrow \mu = \left(\frac{3}{7}, \frac{6}{7}, \frac{2}{7}\right)$$

$$\frac{\partial f}{\partial x} = y + z$$

$$\frac{\partial f}{\partial y} = x + z \rightarrow -1 + 1 = 0$$

$$\frac{\partial f}{\partial z} = x + y$$

$$D_u f(3,6,2) = 1 \cdot \frac{3}{7} + 3 \cdot \frac{6}{7} + 0 \cdot \frac{2}{7} = \frac{3}{7} + \frac{18}{7} = \frac{21}{7} = 3$$



$$e) f(x, y, z) = \sqrt{xyz}, P = (3, 2, 6), \mu = (-1, -2, 2)$$

$$\frac{\partial f}{\partial x} = \frac{1}{2} \cdot (xyz)^{-1/2} \cdot yz = \frac{yz}{2\sqrt{xyz}} \rightarrow \frac{2 \cdot 6}{2\sqrt{3 \cdot 2 \cdot 6}} = \frac{12}{2\sqrt{36}} = \frac{12}{2 \cdot 6} = 1$$

$$\frac{\partial f}{\partial y} = \frac{1}{2} \cdot (xyz)^{-1/2} \cdot xz = \frac{xz}{2\sqrt{xyz}} \rightarrow \frac{3 \cdot 6}{12} = \frac{3}{2}$$

$$\frac{\partial f}{\partial z} = \frac{1}{2} \cdot (xyz)^{-1/2} \cdot xy = \frac{xy}{2\sqrt{xyz}} \rightarrow \frac{3 \cdot 2}{12} = \frac{1}{2}$$

$$\nabla f = (1, 3/2, 1/2) \quad \|\mu\| = \sqrt{(-1)^2 + (-2)^2 + 2^2} = \sqrt{9} = 3 \rightarrow \mu = (-1/3, -2/3, 2/3)$$

$$Du f(-1, -2, 2) = 1 \cdot \left(\frac{-1}{3}\right) + \frac{3}{2} \cdot \left(\frac{-2}{3}\right) + \frac{1}{2} \cdot \frac{2}{3} = \frac{-1}{3} - \frac{6}{6} + \frac{2}{6} = \frac{-1 - 6 + 2}{6} = \frac{-5}{6}$$

9) a) $\frac{df}{dv}(1, -1)$ é mínima $f(x, y) = x^2 - xy + y^2 - y$

$$\frac{\partial f}{\partial x} = 2x - y \quad \frac{\partial f}{\partial y} = -x + 2y - 1$$

aplicando em $(1, -1)$: $\frac{\partial f}{\partial x} = 2 \cdot 1 - (-1) = 3$ $\frac{\partial f}{\partial y} = -1 + 2 \cdot (-1) - 1 = -4$

$$\nabla f(1, -1) = (3, -4)$$

$$V = \nabla f(1, -1) = (3, -4) \quad \|V\| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5 \quad V = (3/5, -4/5)$$

$$\frac{df}{dv}(1, -1) = \|\nabla f(1, -1)\| = 5$$

As direções V que maximizam $D_v f(1, -1)$ são os múltiplos positivos de $(3, -4)$ e o valor máximo da derivada é 5.

b) $\frac{df}{dv}(1, -1)$ é mínima $f(x, y) = x^2 - xy + y^2 - y$

$$\nabla f = (3, -4)$$

$$V = -\nabla f(1, -1) = (-3, 4) \quad \|V\| = \sqrt{(-3)^2 + 4^2} = 5 \quad V = (-3/5, 4/5)$$

$$\frac{df}{dv}(1, -1) = -\|\nabla f(1, -1)\| = -5$$

direção de crescimento min $(3, -4)$

$$\text{valor min} = -5$$

$$c) df/dv (1, -1) = 0$$

$$\nabla f(1, -1) = (3, -4) \quad \parallel df/dv (1, -1) \cdot v = 0 \parallel 3v_1 - 4v_2 = 0 \rightarrow 3v_1 = 4v_2 \rightarrow v_1 = \frac{4}{3}v_2$$

$$v = \left(\frac{4}{3}v_2, v_2\right) = v_2 \left(\frac{4}{3}, 1\right)$$

$$\| \left(\frac{4}{3}, 1\right) \| = \sqrt{\left(\frac{4}{3}\right)^2 + 1^2} = \sqrt{\frac{16}{9} + 1} = \frac{5}{3}$$

$$u = \left(\frac{4}{5}, \frac{3}{5}\right)$$

direções v com derivada direcional zero, todos os vetores não nulos da forma $v = k \left(\frac{4}{3}, 1\right)$, $k \neq 0$.

$$d) df/dv (1, -1) = 4$$

$$\nabla f(1, -1) = (3, -4) \quad df/dv (1, -1) = \nabla f(1, -1) \cdot v = 4$$

$$3v_1 - 4v_2 = 4 \rightarrow -4v_2 = 4 - 3v_1 \rightarrow -4v_2 = 4 - 3v_1 \rightarrow v_2 = (3v_1 - 4)/4$$

direções v com derivada direcional igual a 4: todos os vetores não nulos que satisficem $3v_1 - 4v_2 = 4$

$$e) df/dv (1, -1) = -3$$

$$df/dv \cdot (1, -1) = -3$$

$$\nabla f(1, -1) = (3, -4) \rightarrow df/dv (1, -1) = \nabla f(1, -1) \cdot v = -3$$

$$3v_1 - 4v_2 = -3 \rightarrow -4v_2 = -3 - 3v_1 \rightarrow v_2 = (3v_1 + 3)/4$$

$$10) f(x, y) = (x^2 - y^2)/(x^2 + y^2) \quad p(1, 1)$$

$$\frac{\partial f}{\partial x} = \frac{(x^2 + y^2) \cdot 2x - (x^2 - y^2) \cdot 2x}{(x^2 + y^2)^2} = \frac{2x^2 + 2xy^2 - 2x^3 + 2xy^2}{(x^2 + y^2)^2} = \frac{2xy^2}{(x^2 + y^2)^2} \rightarrow \frac{2 \cdot 1 \cdot 1}{(1^2 + 1^2)^2} = \frac{2}{4} = \frac{1}{2}$$

$$\frac{\partial f}{\partial y} = \frac{(x^2 + y^2) \cdot (-2y) - (x^2 - y^2) \cdot 2y}{(x^2 + y^2)^2} = \frac{-2x^2y - 2y^3 - 2x^2y + 2y^3}{(x^2 + y^2)^2} = \frac{-4x^2y}{(x^2 + y^2)^2} \rightarrow \frac{-4 \cdot 1^2 \cdot 1}{(1^2 + 1^2)^2} = \frac{-4}{4} = -1$$

$$\nabla f(1, 1) \cdot v = 0$$

$$(1, -1) \cdot (v_1, v_2) = 0 \quad v_1 - v_2 = 0 \quad v_1 = v_2$$

$$v = k(1, 1) \quad k \neq 0$$

As direções v em que a derivada direcional de f no ponto $(1, 1)$ é zero são todos os múltiplos não nulos de $(1, 1)$